

Breaking Cross-modal Alignment in Embodied Intelligence: A Multimodal Adversarial Attack Framework for Vision-Language-Action Models

Zhihui Zhao*
College of Computer
Science and Technology
(College of Data Science)
Taiyuan University of
Technology
Taiyuan, Shanxi, China
State Key Laboratory of
Internet Architecture
Tsinghua University
Beijing, China
zhaozihui@tyut.edu.cn

Xiaorong Dong*
College of Computer
Science and Technology
(College of Data Science)
Taiyuan University of
Technology
Shanxi Key Laboratory of
Industrial Internet Security
Taiyuan, Shanxi, China
2023510381@link.tyut.edu.cn

Yaowen Zheng
Institute of Information
Engineering
Chinese Academy of
Sciences
Beijing, China
zhengyaowen@iie.ac.cn

Xiaohui Chen
China Mobile Research
Institute
Beijing, China
chenxiaohuiy@chinamobile.com

Yimo Ren
Institute of Information
Engineering
Chinese Academy of
Sciences
Beijing, China
renyimo@iie.ac.cn

Hangbei Cheng
College of Computer
Science and Technology
(College of Data Science)
Taiyuan University of
Technology
Taiyuan, Shanxi, China
2023510417@link.tyut.edu.cn

Yongle Chen†
College of Artificial
Intelligence
Taiyuan University of
Technology
Shanxi Key Laboratory of
Industrial Internet Security
Taiyuan, Shanxi, China
chenyongle@tyut.edu.cn

Limin Sun
Institute of Information
Engineering
Chinese Academy of
Sciences
Beijing, China
sunlimin@iie.ac.cn

Abstract

Vision-Language-Action (VLA) models underpin robotic and other embodied agents by mapping visual observations and language instructions into executable actions. Their wide adoption through open web model repositories, however, introduces new supply-chain risks: adversaries can launch adversarial attacks to manipulate the action outputs of VLAs, potentially leading to harmful real-world outcomes for embodied agents. To exploit this vulnerability, we propose MAVLA, a novel multimodal adversarial attack framework. MAVLA serves as a modular front-end that integrates seamlessly with a target VLA model, injecting perturbations into task-relevant and structure-sensitive image regions to disrupt cross-modal alignment and induce deviations in the generated action instructions. To balance attack effectiveness with stealth, we design four loss functions that jointly maximize multimodal misalignment while preserving visual stealthiness. Extensive evaluations in simulated and real-world scenarios show that at a 40% perturbation

ratio, the task success rate of VLAs drops by about 70%. Compared to conventional attack baselines, MAVLA achieves superior attack effectiveness and stealthiness with low overhead. Our work reveals a practical and previously underexplored threat to embodied systems, and offers a red-team baseline to inform future defensive strategies and promote safer VLA deployment.

CCS Concepts

• **Computing methodologies** → **Adversarial learning**; • **Security and privacy** → **Vulnerability management**; • **Computer systems organization** → **Robotics**.

Keywords

Embodied Intelligence, Model Supply Chain Security, VLA Model Security, Adversarial Attack, Cross-modal Perturbation

*Both authors contributed equally to this research.

†Corresponding author.



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1 Introduction

Vision-Language-Action (VLA) models have become a core foundation of embodied intelligence [6]. They are applied in robots and other embodied agents. Unlike traditional Vision-Language (VL) models [19, 32], VLA models can jointly encode visual and language modalities and directly generate executable action instructions. This enables a closed loop of “perception–decision–action”. The development and deployment of VLA models rely heavily on the Web ecosystem. Large-scale multimodal training data are mainly collected from open Web platforms. In practice, users often download and reuse pretrained models from open-source communities and model-sharing platforms, and rely on third parties for task-specific fine-tuning [14]. This open web distribution greatly accelerates the adoption of VLAs, but it also introduces new attack surfaces in the model supply chain. Attackers can upload poisoned VLA models to public platforms, which may then be widely downloaded and deployed without awareness of the risks, as shown in Figure 1.

Key Insight. Security of VLA models exhibits a threefold effect. First, multimodal fusion and high-dimensional representations create complex, hard-to-interpret decision boundaries [18, 34]. Second, attackers can apply small, carefully crafted perturbations via visual, language, or cross-modal channels to alter the final action instruction. Third, the outputs of VLA models are mapped to physical actions by embodied agents. Unlike traditional web-supply-chain attacks, a stealthy poisoning event can produce real-world harm to people and environments. For example, a home robot might misclassify a glass as a plastic cup after a tiny image perturbation, causing a failed grasp that breaks the glass or spills liquid. Open Web model platforms make it urgent to systematically study adversarial methods targeted at VLA models. Current work remains scarce in this area. **We focus on a central doubt:** *Can an adversary stealthily manipulate VLA models via poisoning or other adversarial means so that embodied agents behave incorrectly in real-world scenarios?*

Although there is extensive prior work on adversarial attacks for VL models [3, 10, 31, 33], they do not transfer directly to VLA models whose primary output is executable actions. These attack methods fall into three main categories:

- **Pixel-level attacks.** These perturb visual or textual input signals to mislead the model [17, 28]. VLA models rely on multimodal alignment and fusion. Perturbing a single modality often fails to corrupt the fused representation reliably. Such attacks also generalize poorly to unseen scenes, limiting their practical effectiveness [24].
- **Feature-level attacks.** These manipulate intermediate features or hidden representations [10, 29]. In principle, they are more transferable and covert. In practice, however, the mechanisms for perturbing visual-encoder intermediate layers are underexplored. This limits their ability to disrupt cross-modal fusion and to directly alter action generation.
- **Model-manipulation attacks.** These implant backdoors by changing model parameters or architectures [5, 23]. Such attacks usually require white-box access to weights or structure. This requirement is rarely met in real-world deployment, making this class of attacks less practicable.

Method. These works have focused on perturbing a single visual modality. However, because VLA models are trained to closely

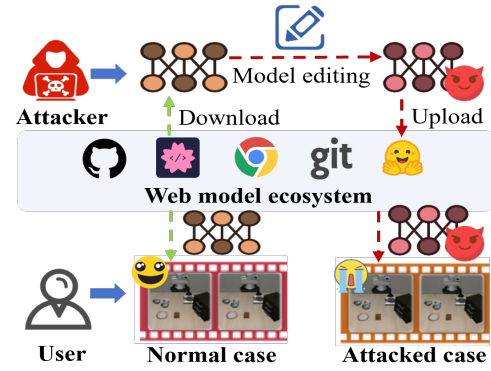


Figure 1: Security Threat to VLA Models in Web Ecosystem

follow textual instructions [13, 25], modifying only the image often fails to mislead the model. Stronger visual perturbations can succeed, but they typically sacrifice stealth. To overcome these limitations, we aim to leverage textual information to guide targeted modifications of image features. We alter visual features so that, given the accompanying textual input, the model can no longer locate the corresponding regions in the image. This effectively disrupts vision–language alignment. Building on this idea, we propose and systematically evaluate a multimodal adversarial attack framework for VLA models, called **MAVLA**. MAVLA designs and inserts a cross-modal perturbation generator between the inputs and the target VLA model. The generator degrades the alignment between vision and language representations at the feature level. Thus, it induces erroneous or potentially dangerous action instructions in downstream embodied agents. To balance attack effectiveness and stealthiness, MAVLA concentrates perturbations on semantically relevant and structure-sensitive regions of the images. This strategy maximizes disruption of multimodal fusion while minimizing human perceptibility.

Evaluation. We conducted extensive experiments to validate the attack efficacy of MAVLA. In the **simulation**, we conducted evaluations using the LIBERO-Spatial and LIBERO-Goal datasets [16] on the target models, OpenVLA [12] and CEED-VLA [26]. In the **real-world scenario**, we deployed a 6-DoF robotic arm integrated with a fine-tuned OpenVLA model. We measured the task success rates of the VLAs under different MAVLA attack intensities. Results show that in simulation, when the perturbation ratio reached 40%, the success rate dropped about 70%, indicating a near-complete degradation of the models’ fundamental capabilities. Visualization analysis helped identify the optimal configuration of attack intensity (i.e., perturbation ratio $\approx 25\%$). We compared MAVLA against common adversarial methods in VL models—such as Random Noise, Frequency Noise, and Local Gaussian Mask—in terms of both attack effectiveness and perturbation stealthiness. The results show that MAVLA consistently outperforms these baselines in both metrics. The additional overhead from MAVLA was evaluated, measuring task Time-to-Completion and deployment latency. Results indicated an inference time increase of a mere 4ms and the increase in model parameter count is about 0.6%. In real-world scenarios, the task success rate decreased by 26.8% following the attack.

Contributions. Our contributions are outlined as follows.

- MAVLA, a multimodal adversarial attack framework, functions as a modular front-end to break cross-modal alignment in VLA models, thereby exposing a new attack surface.
- MAVLA concentrates perturbations on semantically relevant and structure-sensitive regions of the image, thereby maximizing disruption to multimodal fusion while minimizing human perceptibility.
- This work demonstrates MAVLA’s effective and stealthy attacks through extensive simulated and real-world evaluations, thereby exposing a critical supply-chain vulnerability in VLAs sourced from open-web platforms and highlighting the need for safer deployment practices.
- We release the source code of MAVLA on <https://github.com/Dongxiaorong/0607/MAVLA> to facilitate reproducibility and further studies.

2 Threat Model and Attack Objectives

In this paper, our adversary aims to manipulate the action decisions in VLA models through the injection of adversarial vision perturbations. We consider a practical **semi-black-box attacker** who has knowledge of the model architecture and pretrained weights (e.g., obtained from public repositories) but lacks access to the original pretraining corpus or downstream fine-tuning datasets [34]. An attacker can execute the adversary attacks for VLA models and their applications, which closely aligns with model supply chain security concerns. We make the following **assumptions** about the attacker’s capabilities and requirements (i.e., threat model):

- Attacker can query the VLA model and observe its outputs, including predicted action sequences or trajectories.
- Attacker cannot access ground-truth action labels or downstream task datasets but may optionally access intermediate visual features (e.g., attention maps, value vectors) during offline analysis or debugging.
- Attacker cannot modify model weights or training data. It can insert perturbation modules between the input multimodal data and the model.
- Perturbations must be imperceptible to human observers, are generated without downstream task supervision and must avoid obvious visual or semantic artifacts.

Based on this threat model, the adversary seeks to produce perturbations that satisfy three practical **attack objectives**:

- **High effectiveness.** Cause large, meaningful deviations in embodied agent behaviors. Examples include incorrect object manipulations, execution of unsafe or unintended actions, or total task failure in embodied systems.
- **Feature-level transferability.** The perturbation is applied to the visual features produced by the vision encoder, not to raw image pixels. It remains effective across diverse downstream tasks and heterogeneous VLA model architectures. That is, the attack should generalize, rather than being confined to the tasks of a specific VLA model.
- **Stealthiness.** Make perturbations minimally affect the visual or text contents, ensuring that they remain imperceptible and preserve input fidelity, while maintaining the effectiveness in compromising the target VLA model.

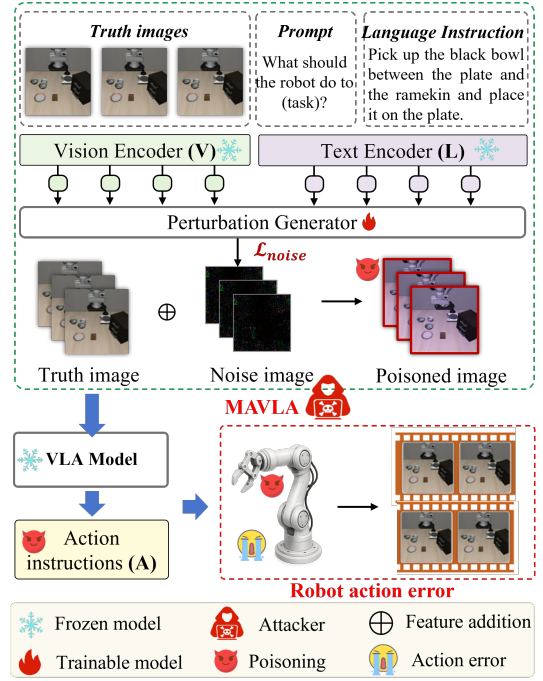


Figure 2: Overview of MAVLA

3 Methodology

We propose MAVLA, a modular adversarial front-end designed to realize the attack objectives. MAVLA first re-uses the target VLA model’s vision and text encoders to obtain the visual and textual features of the inputs. Using these encoded features, we train a well-designed perturbation generator that serves as the core component of MAVLA. MAVLA operates as an independent pre-processing module that is inserted before the non-encoder parts of target VLA model, as illustrated in Figure 2. When integrated with the target VLA’s downstream components, MAVLA’s attack pipeline produces a poisoned VLA model, resulting in incorrect instructions and unsafe physical actions in the embodied agents.

Concretely, given an image–text input pair (I, T) , the I is mapped to visual features and the T to text features by MAVLA’s encoders. These extracted features are subsequently fed into the MAVLA’s perturbation generator, and guide to produce an imperceptible perturbation at specific locations of images. The feature-level manipulation causes the downstream policy to produce incorrect instructions. Perturbation generator locates in **Section 3.1**.

Feature encoder extraction from VLA models is a common practice. Our focus lies in how the encoded features can guide the construction and training of a perturbation generator to achieve our attack objectives, particularly through the loss functions employed during its training. Specifically, to achieve both attack effectiveness and feature-level transferability, we maximize two discrepancies: the first, called the *feature misleading loss*, is the gap between clean and perturbed features; the second, the *alignment misleading loss*, is the gap between image features and the semantics conveyed by text features. By modifying only the image features, maximizing these

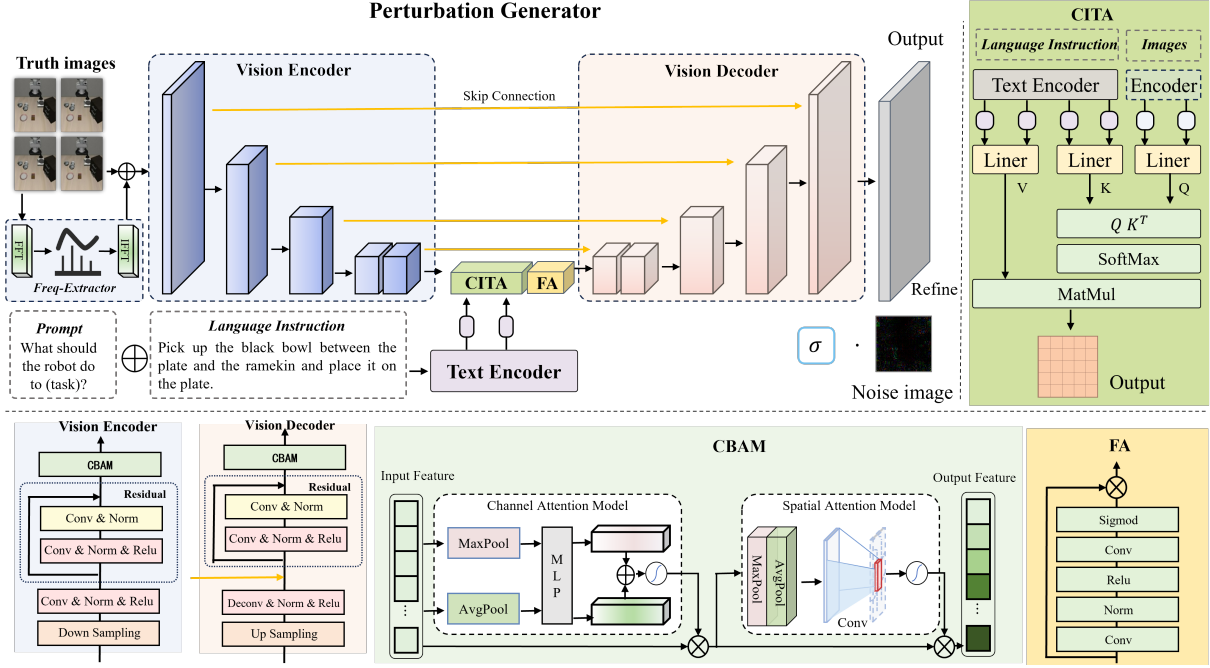


Figure 3: The Network of Perturbation Generator

two losses ensures that adversarial samples preserve semantic consistency while misleading cross-modal representations. To ensure stealthiness, we minimize both the structural difference between the original and perturbed images (i.e., the *structure-aware loss* [15]) and the perturbation intensity (referred to as the *perturbation constraint loss*). This enables the adversarial samples to remain nearly indistinguishable to humans without significantly compromising attack effectiveness. These loss functions are introduced in detail in **Section 3.2**.

3.1 Perturbation Generator

The network of perturbation generator G_θ is shown as Figure 3. The generator takes as input an image $x \in \mathbb{R}^{B \times C \times H \times W}$ and a task embedding T , and outputs a residual-style perturbation $\delta = G_\theta(x, T)$. The generator network consists of three components: *Vision Encoder-Decoder Backbone*, *Cross-modal Instruction-Token Attention (CITA)* module embedded at the bottleneck, and *Frequency-domain Augmentation (FA)* module.

3.1.1 Vision Encoder-Decoder Backbone. The Vision Encoder-Decoder backbone adopts a simplified U-Net [22] with symmetric encoder and decoder branches. The encoder progressively downsamples to extract multi-scale semantic features, while the decoder gradually upsamples to reconstruct perturbations. Finally, a refinement module constrains the perturbation amplitude.

In the encoder, each encoder stage performs downsampling with a stride-2 convolution, followed by normalization, activation, a residual block, and CBAM attention:

$$E_{l+1} = \text{CBAM}(\text{Res}(\text{ReLU}(\text{BN}(\text{Conv}^{s=2}(E_l)))))) \quad (1)$$

In the decoder, each decoder stage upsamples features using transposed convolution (or interpolation + convolution), fuses them with the corresponding encoder features via skip connections, and then applies residual and CBAM modules:

$$\widehat{D}_l = \text{ReLU}(\text{BN}(\text{Up}(D_{l+1}))), \quad D_l = \text{CBAM}(\text{Res}(\widehat{D}_l \oplus E_l)) \quad (2)$$

where $\oplus$ denotes element-wise addition.

For the output, residual blocks and convolutions are applied, followed by a $\tan h$ activation to constrain the perturbation amplitude:

$$\delta = \tan h(f_{\text{refine}}(D_1)) \quad (3)$$

To mitigate feature loss during image encoding, we integrate three key components: Skip Connections [35] preserve high-resolution details and alleviate information loss during downsampling. Residual Blocks [27] stabilize deep network training and improve convergence CBAM Attention Mechanism [30] adaptively emphasizes informative channels and spatial regions that exhibit strong adversarial potential.

3.1.2 Cross-modal Instruction-Token Attention. To achieve a deep integration of visual features and text semantics, we design an CITA module, where the *Cross-Attention* branch injects text semantics into visual features for selective modulation, and the *Multi-Modal Self-Attention* branch models global dependencies on the enhanced representation. Thus, both improve spatial expressiveness while maintaining cross-modal consistency.

In the CITA module, the input visual feature comes from the last encoder stage, denoted as $\mathbf{V} \in \mathbb{R}^{B \times 512 \times H/16 \times W/16}$, and the task-related text embedding is represented as $\mathbf{t} \in \mathbb{R}^{B \times D}$, where B denotes the batch size and D is the text embedding dimension. To align the

modalities, the text embedding is first projected into the visual feature space via a learnable matrix $\mathbf{W}_p \in \mathbb{R}^{512 \times D}$:

$$\mathbf{t}' = \mathbf{W}_p \mathbf{t}, \quad \mathbf{t}' \in \mathbb{R}^{B \times 512} \quad (4)$$

The visual feature map is then flattened into a sequence $\mathbf{V}_{flat} \in \mathbb{R}^{B \times (H/16 \cdot W/16) \times 512}$ to generate query vectors (Q), while the projected text embedding is mapped into key (K) and value (V) vectors:

$$\mathbf{Q} = \mathbf{V}_{flat} \mathbf{W}_q, \quad \mathbf{K} = \mathbf{t}' \mathbf{W}_k, \quad \mathbf{V}_t = \mathbf{t}' \mathbf{W}_v \quad (5)$$

where $\mathbf{W}_q, \mathbf{W}_k, \mathbf{W}_v \in \mathbb{R}^{512 \times d_h}$ are learnable matrices and d_h is the hidden dimension. The attention weights are computed as:

$$\mathbf{A} = \text{Softmax}\left(\frac{\mathbf{Q}\mathbf{K}^T}{\sqrt{d_h}}\right) \quad \mathbf{A} \in \mathbb{R}^{B \times (H/16 \cdot W/16) \times 1} \quad (6)$$

The cross-attended representation is obtained by weighting the text values and fusing them with the original visual features via residual connection:

$$\mathbf{V}' = \mathbf{V} + \mathbf{A} \mathbf{V}_t \mathbf{W}_o, \quad \mathbf{V}' \in \mathbb{R}^{B \times 512 \times H/16 \times W/16} \quad (7)$$

By using semantic guidance to highlight task-relevant regions, the process not only enhances discriminative feature representation but also preserves training stability through residual fusion.

After obtaining the cross-modally enhanced features, the enhanced feature map $\mathbf{V}'$ is flattened into a sequence and processed by a multi-head attention mechanism, where each head generates its own queries, keys, and values. Within each head, the model computes the pairwise correlations between any positions in the sequence and aggregates contextual information globally:

$$\text{Attn}(\mathbf{X}) = \text{Softmax}\left(\frac{\mathbf{X}\mathbf{W}_q(\mathbf{X}\mathbf{W}_k)^T}{\sqrt{d_h}}\right)(\mathbf{X}\mathbf{W}_v) \quad (8)$$

The fused representation from the multi-head concatenation as:

$$\widehat{\mathbf{V}} = f_{\text{MMA}}(\mathbf{V}, \mathbf{t}), \quad \widehat{\mathbf{V}} \in \mathbb{R}^{B \times 512 \times H/16 \times W/16} \quad (9)$$

3.1.3 Frequency Domain Augmentation. We introduce the FA module to generate covert and controllable perturbations that avoid the visible artifacts typical of direct spatial manipulation. Specifically, this is achieved by transforming the image into the frequency domain via 2D Fast Fourier Transform (FFT) [4]; suppressing low-frequency bands while retaining high-frequency components rich in edge and texture information; and then transforming the masked spectrum back via inverse FFT (iFFT) [2]. The resultant signal is residually fused with the original image, enhancing fine details without perceptual distortion.

Formally, the 2D FFT of x is written as:

$$X(u, v) = \mathcal{F}\{x\}(u, v) = \sum_{m=0}^{H-1} \sum_{n=0}^{W-1} x(m, n) e^{-2\pi i \left(\frac{um}{H} + \frac{vn}{W} \right)} \quad (10)$$

With cutoff ratio $c \in (0, 1)$, we define

$$h_c = \lfloor H \cdot c \rfloor, \quad w_c = \lfloor W \cdot c \rfloor \quad (11)$$

and construct a high-frequency mask by zeroing the central low-frequency region. After inverse FFT, we obtain

$$F(x) = \mathcal{R}\{\mathcal{F}^{-1}(\widetilde{X})\} \in \mathbb{R}^{B \times C \times H \times W} \quad (12)$$

where $\widetilde{X}$ is the masked spectrum. Then, the frequency-enhanced signal is injected as

$$x' = x + \alpha F(x) \quad (13)$$

where α is a scaling factor. This step highlights discriminative high-frequency cues while maintaining global shape consistency.

3.2 Loss Functions

In these loss functions, to ensure the stealthiness of adversarial perturbations, we adopt a perturbation constraint loss $\mathcal{L}_{noise}$ to limit its magnitude, and introduce a structure-aware loss $\mathcal{L}_{lpips}$ to maintain the integrity of the original image. Meanwhile, to enhance the attack effectiveness, we utilize a feature misleading loss $\mathcal{L}_{feature}$ to maximize the discrepancy between original and perturbed features, and an alignment misleading loss $\mathcal{L}_{alignment}$ to induce misalignment between the image and text. Four loss functions jointly ensure that the perturbations remain minimal and the attack impact is maximized. The optimization objective as:

$$\mathcal{L}_{total} = \lambda_1 \mathcal{L}_{noise} + \lambda_2 \mathcal{L}_{feature} + \lambda_3 \mathcal{L}_{alignment} + \lambda_4 \mathcal{L}_{lpips} \quad (14)$$

where λ_i denotes the weight for each loss function.

3.2.1 Perturbation Constraint Loss. Perturbation constraint loss directly constrains the perturbation tensor δ . Specifically, we apply both L_2 regularization and Total Variation (TV) regularization:

$$\mathcal{L}_{noise} = \|\delta\|_2^2 + \beta \cdot TV(\delta) \quad (15)$$

The L_2 norm and the Total Variation (TV) term, balanced by β , collectively enforce stealthiness: the former limits the global perturbation strength, and the latter penalizes local intensity variations to discourage noisy artifacts. This ensures the adversarial example is visually indistinguishable from the original.

3.2.2 Feature Misleading Loss. Feature misleading loss should maximize the discrepancy between clean and perturbed samples in the feature space. This loss consists of two complementary components.

Global difference. We measure the cosine similarity between the global feature embeddings of the clean and perturbed samples, and enforce them to diverge:

$$\mathcal{L}_{global} = 1 - \cos(f(I), f(I + \delta)) \quad (16)$$

where $f(\cdot)$ denotes the vision encoder. By decreasing the cosine similarity, we ensure that perturbations induce significant deviations at the global semantic level.

Local structural difference. We leverage token-level features extracted from multiple Transformer layers, and compute both mean squared error (MSE) [8] and Gram matrix discrepancy:

$$\mathcal{L}_{local} = \frac{1}{L} \sum_{l=1}^L \left(\|F_l(I) - F_l(I + \delta)\|_2^2 + \|G(F_l(I)) - G(F_l(I + \delta))\|_2^2 \right) \quad (17)$$

where $F_l(\cdot)$ represents the feature map at the l -th layer, and $G(\cdot)$ denotes the Gram matrix that captures pairwise correlations among feature channels. This term enforces perturbations to disrupt not only local activations but also higher-order feature correlations. The feature misleading loss is a weighted as:

$$\mathcal{L}_{feature} = \alpha \mathcal{L}_{local} + (1 - \alpha) \mathcal{L}_{global} \quad (18)$$

where α controls the balance between local and global misguidance.

3.2.3 Alignment Misleading Loss. A challenge in attacking VLA models lies in disrupting cross-modal alignment, i.e., the correspondence between images and textual instructions. We propose an alignment misleading loss based on a relative mismatch strategy. Let v_{ori} and v_{pert} denote the visual features of the clean and perturbed images, respectively, and let t denote the text feature. Using cosine similarity $\text{sim}(\cdot, \cdot)$, the loss is defined as:

$$\mathcal{L}_{\text{alignment}} = \max(0, \text{sim}(v_{pert}, t) - \text{sim}(v_{ori}, t)) \quad (19)$$

This function directly reduces the alignment of the perturbed image-text pair to mislead multimodal reasoning, where even slight perturbations can induce a significant drop in perceived instruction relevance, triggering incorrect action predictions.

3.2.4 Structure-aware Loss. To prevent the degradation of perceptual quality and enhance stealthiness, we introduce a Structure-aware Loss. This loss employs the Learned Perceptual Image Patch Similarity (LPIPS) metric:

$$\mathcal{L}_{\text{lips}} = \text{LPIPS}(I, I + \delta) \quad (20)$$

LPIPS measures similarity in a deep perceptual feature space rather than pixel space, providing a guarantee of semantic preservation. By minimizing this loss, the perturbed images remain perceptually indistinguishable from clean ones, improving attack stealthiness.

4 Experiments

4.1 Experimental Settings

Simulation Scenario. We employ the OpenVLA [12] as the victim VLA. To examine MAVLA’s feature-level transferability, we additionally test its effectiveness on CEED-VLA [26]. We conduct experiments on the LIBERO benchmark [16], a large-scale suite of embodied intelligence tasks designed for evaluating VLA models. Due to our device resource constraints, this paper focuses on two datasets: LIBERO-Spatial [21] and LIBERO-Goal [11], covering a total of 20 diverse household and manipulation tasks. Specifically, LIBERO-Spatial consists of 10 tasks that emphasize variations in the spatial positions of objects. Meanwhile, LIBERO-Goal contains 10 tasks that focus on variations in task goals, requiring the agent to achieve distinct target states or perform different actions.

Real-world Scenario. We adopt a robotic platform consisting of a Yahboom AI Vision Programming Robotic Arm with six degrees of freedom (6DoF) and a parallel gripper for grasping operations. The sensing system includes a fixed-position RGB camera with a shoulder-view perspective. In this setting, a fine-tuned OpenVLA model is deployed to control the robotic arm. Inspired by the LIBERO benchmark, we design and implement grasping tasks on a private real-world dataset. During each task execution, a high-definition camera was used to capture third-person view images of the robotic arm, while its pose data were simultaneously recorded. This setting ensures both reproducibility of tasks and completeness of multimodal features, as shown in Figure 4.

Attacks. We compare our proposed MAVLA against representative baseline adversarial attacks:

- **Random Noise [1].** This baseline directly injects random Gaussian noise into the input image without optimization or semantic constraints, serving as an unstructured and task-agnostic perturbation.

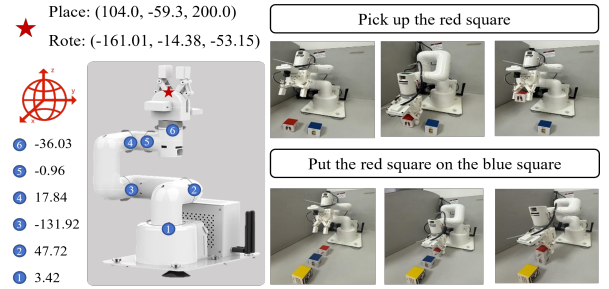
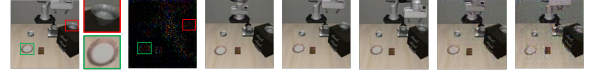
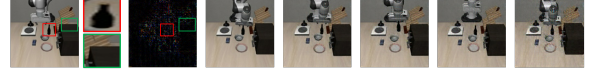


Figure 4: Private Real-world Dataset

LIBERO-Spatial: pick up the black bowl on the wooden cabinet and place it on the plate.



LIBERO-Goal: put the wine bottle on top of the cabinet.



Original Perturbed $\sigma = 0.1$ $\sigma = 0.2$ $\sigma = 0.25$ $\sigma = 0.3$ $\sigma = 0.4$

Figure 5: Visualization results under different σ

- **Local Gaussian Mask [20].** A smooth, localized perturbation is applied at a specific position in the image and weighted by a Gaussian function. The perturbation intensity gradually decays toward the boundary, creating a natural transition and enhancing stealthiness to observers.
- **Frequency Noise [9].** Convert the image to grayscale, apply a centered 2D DFT, preserve the central low-frequency band and replace other coefficients with Gaussian noise, inverse-DFT and min-max normalize, broadcast to three channels and mean-center the perturbation.

Metrics. We evaluate MAVLA from three perspectives:

- **Attack Effectiveness and Transferability.** We measure the *Task Success Rate drop (TSR)* on LIBERO-Spatial dataset, LIBERO-Goal dataset and real-world tasks. *TSR* quantifies the degradation of task-level performance. It captures attack effectiveness and feature-level transferability.
- **Perturbation Stealthiness.** We quantify perturbation magnitude using L_p -norm constraints and measure perceptual similarity with *LPIPS* [7]. The L_p -norm provides a formal bound on the perturbation budget, while *LPIPS* captures perceptual differences that go beyond pixel-wise metrics. These measure evaluates the attack’s stealthiness.
- **Computational and Deployment Cost.** We assess the MAVLA’s runtime overhead and resource requirements. The metrics include *Time-to-Completion (TTC)*, *extra deployment latency (EDL)*, *peak memory usage (PM)*, and *attack generation time (AGT)*. We also report the change in the VLA model’s parameter count (*MP*) induced by MAVLA.

Table 1: TSR of Two VLA Models under Different σ

Datasets	Model	σ	Task 1	Task 2	Task 3	Task 4	Task 5	Task 6	Task 7	Task 8	Task 9	Task 10	Total
LIBERO-Spatial	OpenVLA	0.00	82.4%	94.7%	88.2%	90.5%	74.6%	80.3%	96.3%	90.1%	84.8%	74.2%	85.6%
		0.10	76.2%	88.5%	90.1%	96.3%	54.1%	88.3%	97.5%	84.2%	74.5%	68.1%	81.8%
		0.20	72.2%	77.8%	72.4%	91.7%	58.2%	87.9%	86.2%	28.9%	82.2%	46.4%	70.4%
		0.25	68.2%	73.8%	58.3%	92.7%	46.5%	64.6%	72.2%	18.9%	73.8%	21.7%	59.1%
		0.30	62.4%	62.1%	26.8%	75.1%	26.1%	12.5%	32.1%	24.8%	63.4%	8.1%	39.3%
		0.40	38.6%	18.1%	38.4%	2.9%	4.3%	6.7%	0.0%	30.9%	2.0%	20.5%	16.2%
	CEED-VLA	0.00	100.0%	84.5%	87.9%	95.1%	82.3%	92.1%	86.4%	77.6%	86.4%	73.9%	86.6%
		0.10	92.8%	85.9%	78.2%	93.8%	66.9%	75.7%	85.2%	90.1%	90.7%	69.2%	82.9%
		0.20	90.8%	100.0%	70.1%	87.8%	57.4%	76.8%	71.8%	88.3%	91.4%	53.8%	78.8%
		0.25	89.6%	92.6%	74.2%	81.8%	39.1%	92.3%	81.6%	71.6%	86.3%	34.1%	74.3%
		0.30	88.2%	91.2%	69.3%	79.4%	22.6%	94.6%	62.3%	49.8%	84.7%	44.6%	68.7%
		0.40	81.7%	68.8%	45.9%	73.7%	7.9%	79.1%	33.5%	0.0%	85.1%	10.0%	48.6%
LIBERO-Goal	OpenVLA	0.00	54.4%	88.1%	85.9%	61.5%	88.2%	73.9%	76.2%	95.9%	84.2%	64.1%	77.2%
		0.10	48.2%	82.3%	84.5%	58.7%	85.3%	75.7%	65.8%	92.1%	81.7%	56.2%	73.1%
		0.20	42.4%	63.6%	69.3%	30.1%	94.2%	65.8%	71.6%	93.8%	70.4%	42.2%	64.3%
		0.25	46.3%	52.1%	66.6%	25.1%	88.1%	53.7%	52.1%	85.8%	72.2%	36.3%	57.8%
		0.30	32.3%	62.2%	67.4%	36.3%	94.4%	66.2%	64.1%	92.5%	69.2%	39.4%	62.4%
		0.40	1.8%	0.0%	17.5%	0.0%	24.1%	4.2%	8.1%	20.7%	0.0%	8.1%	8.4%
	CEED-VLA	0.00	58.2%	84.5%	90.1%	31.7%	99.6%	78.9%	89.5%	96.9%	92.3%	58.0%	78.0%
		0.10	42.4%	78.3%	81.2%	26.1%	97.8%	86.9%	86.5%	96.1%	90.4%	42.2%	72.8%
		0.20	54.2%	68.5%	70.1%	10.1%	98.2%	66.5%	88.2%	96.1%	86.2%	39.1%	67.7%
		0.25	46.2%	32.8%	26.7%	8.2%	86.1%	38.4%	72.3%	76.2%	66.7%	34.5%	48.8%
		0.30	34.7%	16.2%	28.4%	2.1%	56.1%	29.8%	54.2%	54.6%	45.2%	19.5%	34.1%
		0.40	16.2%	2.1%	2.3%	0.0%	28.5%	15.2%	4.8%	2.1%	4.3%	12.1%	8.8%

4.2 Simulation Results

4.2.1 Attack Effectiveness and Transferability. We compare the *TSR* before and after deploying MAVLA, as shown in Table 1. To simulate different attack intensities, we added perturbations at varying ratios (i.e., σ) to the visual features and evaluated the performance of each VLA model across multiple sub-tasks. The results show that as σ increases, the *TSR* consistently decreases. In the OpenVLA and LIBERO-Spatial settings, when σ reaches 40%, some sub-tasks drop to a *TSR* of 0%, thereby showing attack effectiveness. Similar trends are observed in the CEED-VLA and LIBERO-Goal combinations, confirming the feature-level transferability. At this point, the VLA model can be regarded as having nearly lost its ability to perform downstream tasks. Furthermore, we visualized the perturbed images during the attack process, as shown in Figure 5. As σ increases, the perturbations become progressively easier to detect. Since perturbations at $\sigma = 0.4$ are already visually noticeable to human observers, which contradicts our stealthiness objective, we restrict our evaluation to $\sigma \leq 0.4$. We also found that the optimal configuration of attack intensity about $\sigma \approx 0.25$. Notably, MAVLA introduces perturbations at semantically critical regions, thereby amplifying their disruptive impact on local targets. To show this effect, we zoomed in on the targeted regions of the perturbed images.

4.2.2 Perturbation Stealthiness. We evaluate MAVLA’s perturbation stealthiness on OpenVLA across two datasets ($\sigma = 0.25$) against three baseline attacks. The evaluation metrics include *TSR* for attack effectiveness, and L_p -norm and *LPIPS* for perceptual similarity. As detailed in Appendix C (Table 4), MAVLA consistently

achieves superior attack efficacy (significant drop in *TSR*) with the smallest perturbations. The minimal *LPIPS* scores further confirm that MAVLA generates nearly imperceptible perturbations, effectively balancing high attack strength with visual stealthiness. Furthermore, Figure 6 visualizes perturbation features under different attacks. MAVLA’s perturbations are virtually invisible to human observers and, compared with baseline attacks, are more precisely concentrated on local targets, thereby enhancing both attack stealthiness and effectiveness.

Table 2: Impact of Different Combinations of Loss Functions During The Training Phase on Model Performance

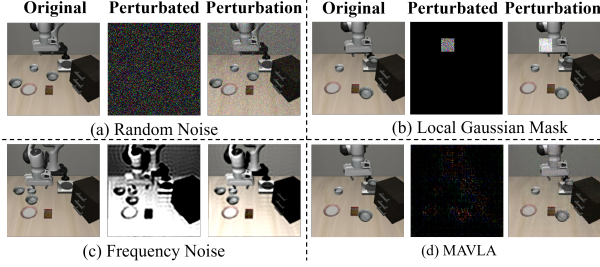
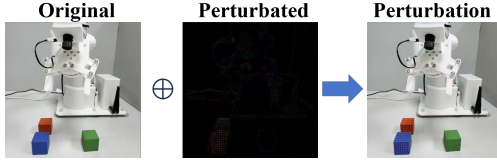
	$\mathcal{L}_f$	$\mathcal{L}_n$	$\mathcal{L}_t$	$\mathcal{L}_a$	LIBERO-Spatial		LIBERO-Goal	
					<i>TSR</i>	<i>LPIPS</i>	<i>TSR</i>	<i>LPIPS</i>
OpenVLA	-	-	-	-	85.6%	0	77.2%	0
	✓	×	×	×	40.4%	5.126	35.6%	4.943
OpenVLA+MAVLA	✓	✓	×	×	64.7%	0.703	70.5%	0.936
	✓	✓	✓	×	65.3%	0.334	69.4%	0.421
	✓	✓	✓	✓	59.1%	0.078	57.8%	0.131

4.2.3 Ablation Study. To validate the attack effectiveness of MAVLA’s training strategy and its individual components, we conducted an ablation study using the OpenVLA model on the LIBERO-Spatial and LIBERO-Goal datasets, with the fixed $\sigma = 0.25$.

We evaluated the impact of different combinations of loss functions during the training on model performance. The $\mathcal{L}_{feature}$

Table 3: Ablation Study of The Perturbation Generator

	CITA	Freq-domain	LIBERO-Spatial		LIBERO-Goal	
			TSR	LPIPS	TSR	LPIPS
OpenVLA	-	-	85.6%	0	77.2%	0
	×	×	55.4%	0.379	60.9%	0.417
OpenVLA+MAVLA	✓	×	63.1%	0.126	62.5%	0.221
	✓	✓	59.1%	0.078	57.8%	0.131

**Figure 6: Visualization of perturbation features****Figure 7: Example Attack Images in The Real-world Scenarios**

($\mathcal{L}_f$) was used as the primary objective, maximizing the discrepancy between perturbed and original features. We then progressively incorporated additional terms. The $\mathcal{L}_{noise}$ ($\mathcal{L}_n$) controlled the magnitude of the generated perturbations, while the $\mathcal{L}_{lips}$ ($\mathcal{L}_l$) enhanced the stealthiness of the perturbations and reduced the likelihood of detection. Finally, the $\mathcal{L}_{alignment}$ ($\mathcal{L}_a$) was introduced to guide the perturbations toward local task-relevant objectives. As shown in Table 2, using only $\mathcal{L}_f$ led to uncontrolled perturbation growth. While the TSR dropped significantly, the resulting perturbations were highly perceptible and inconsistent with our attack objectives. After adding the $\mathcal{L}_n$, TSR improved, and the stealthiness was effectively enhanced. Incorporating $\mathcal{L}_l$ further reduced detectability while maintaining similar attack effectiveness. With the $\mathcal{L}_a$, perturbations became concentrated on task-relevant regions, simultaneously enhancing attack effectiveness and stealthiness.

We further conducted an ablation study on the key components of the perturbation generator by comparing three configurations: (i) no additional component (relying solely on unimodal image features); (ii) incorporating CITA module; and (iii) augmenting (ii) with frequency-domain fusion. The results are summarized in Table 3. Without additional components, TSR decreases by roughly 30% relative to the no-attack baseline; however, $LPIPS$ increases sharply, indicating that the generated perturbations are cluttered and thus more perceptible. With CITA, the perturbations are semantically guided and concentrated around local targets. For example,

on LIBERO-Goal, $LPIPS$ drops to 0.221 while TSR rises slightly by 1.6%, indicating enhanced stealthiness without sacrificing effectiveness. Building on this, integrating frequency-domain fusion further suppresses spurious textures and noise in non-target regions at the spectral level. On LIBERO-Spatial, $LPIPS$ decreases to 0.078. Overall, CITA provides semantic-level target constraints, while frequency-domain fusion offers fine-grained spectral regulation. Their synergy yields more precise and less perceptible perturbations, achieving a balance between stealthiness and attack effectiveness.

4.2.4 Computational and Deployment Cost. We assessed MAVLA’s deployability on OpenVLA (LIBERO-Spatial and LIBERO-Goal) via TTC , EDL , PM , and AGT , with results in Appendix D (see Table 5). While TTC increases by $\sim 50\%$ due to attack-induced corrective actions, the resource footprint is low: one-time setup is negligible ($EDL \leq 0.13s$), PM grows by only 1.5%, and sustained AGT remains under 4 ms (0.6% parameter increase). Overall, MAVLA ensures strong attack effectiveness with controlled overhead suitable for near-real-time deployment.

4.3 Real-world Results

We also evaluated the effectiveness of generated adversarial perturbations in a real-world setting. The evaluation covered two task types: object grasping and object placement, as illustrated in Figure 4. Deploying MAVLA on OpenVLA and the physical robotic arm reduced the task success rate by 26.8%, demonstrating that MAVLA’s perturbations remain effective in real-world scenarios without further adaptation. An attacker could further fine-tune MAVLA to increase attack effectiveness for specific targets or environments. Example attack images are shown in Figure 7.

5 Conclusion

This paper focuses on cross-modal alignment of VLA models which is the primary vulnerability in embodied intelligence and presents MAVLA, a high-efficiency, low-overhead, and transferable adversarial front end. We validate MAVLA’s effectiveness, stealthiness and transferability in both simulation and real-world environments. Results show that MAVLA reduces TSR while preserving stealthiness, showing practical engineering feasibility. Our work follows an offense-to-inform-defense paradigm. MAVLA provides a reusable red-team baseline for VLA security evaluation and a concrete adversary model for designing and testing defenses.

In the future, we will explore defensive countermeasures, including robustness-oriented training for cross-modal alignment, lightweight online detection and isolation, and risk-tiered security evaluation. Ultimately, we aim to establish a systematic “attack first, strengthen later” methodology to accelerate the safe and trustworthy deployment of embodied intelligence.

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A Embodied Intelligence and VLA models

Embodied intelligence represents an important direction in the evolution of artificial intelligence from the information space to the physical world, with typical applications such as robots, quadruped robots, and unmanned aerial vehicles. It emphasizes direct interaction with the real environment and demonstrates a certain degree of autonomous decision-making capability. By forming a perception-action-feedback loop in real or virtual environments, embodied agents can gradually learn and master complex skills such as motor control, task planning, and semantic understanding. This closed-loop mechanism also enables stronger environmental adaptability and cross-task generalization.

VLA models are a core enabler of embodied intelligence. They are designed to jointly process visual perception, natural language instructions, and sequential decision-making, thereby achieving an integrated pipeline of "seeing-understanding-acting".

Suppose the input is a visual observation $I \in \mathbb{R}^{H \times W \times C}$, where H , W , and C denote the height, width, and number of channels of the image, respectively. At the same time, a language instruction is given as a sequence

$$Q = (w_1, w_2, \dots, w_T)$$

where each w_t represents the t -th word and T is the sequence length. The goal of the model is to generate an action sequence $A = (a_1, a_2, \dots, a_K)$, where a_t denotes the t -th action and K is the sequence length.

Vision encoder, denoted by $f_{\text{vision}}(\cdot)$, maps the image I into a visual feature vector $v \in \mathbb{R}^d$, where d is the dimension of the feature space. Similarly, the language encoder $f_{\text{lang}}(\cdot)$ maps the instruction

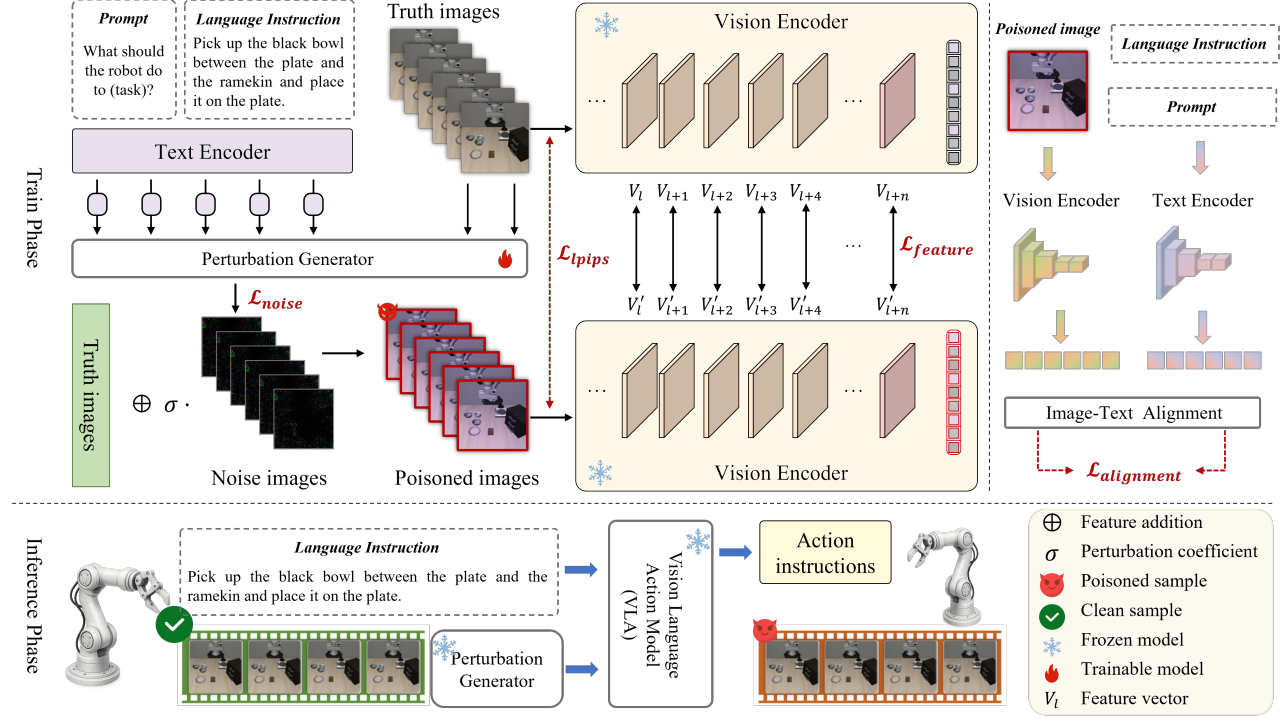


Figure 8: The detailed training phase and four loss functions.

Q into a language feature vector $l \in \mathbb{R}^d$, which shares the same feature space with v to facilitate cross-modal alignment.

Given the visual feature v and the language feature l , the VLA model employs a policy function $f_{\text{policy}}(\cdot)$ to fuse the two modalities and predict the next action a_t , conditioned on the previously generated action history $a_{<t} = (a_1, a_2, \dots, a_{t-1})$. This process can be formalized as:

$$a_t = f_{\text{policy}}(v, l, a_{<t}), \quad t = 1, \dots, K. \quad (21)$$

The training objective is to minimize the discrepancy between the predicted actions and the ground-truth action sequence $A^* = (a_1^*, a_2^*, \dots, a_K^*)$. The corresponding loss function is defined as:

$$\mathcal{L} = - \sum_{t=1}^K \log P(a_t^* | v, l, a_{<t}), \quad (22)$$

where $P(a_t^* | v, l, a_{<t})$ denotes the probability that the model correctly predicts the ground-truth action a_t^* given the visual feature, language feature, and the history of actions.

VLA provides a unified framework for perception, language, and decision-making, aligning visual observations with natural language instructions and generating action sequences consistent with task requirements. In this way, VLA serves as a fundamental approach to achieving embodied intelligence.

B Overview of MAVLA and four loss functions

Figure 8 illustrates the detailed training phase and the specific application of these four loss functions. In the train phase, the perturbation generator is optimized to adversarial perturbation under

strict constraints. Specifically, the perturbation constraint loss (denoted as $\mathcal{L}_{noise}$) is applied directly to the noise map to limit its intensity, while the structure-aware loss (denoted as $\mathcal{L}_{lips}$) measures the perceptual difference between the truth and poisoned images. To realize the attack objectives, MAVLA maximizes the feature misleading loss (denoted as $\mathcal{L}_{feature}$) by calculating the discrepancy between the intermediate feature vectors (V_l and V'_l) of the clean and poisoned streams. Simultaneously, the alignment misleading loss (denoted as $\mathcal{L}_{alignment}$) is employed at the output stage to disrupt the semantic consistency between the image embeddings and the language instructions. The Inference Phase in the bottom panel shows how this trained module is integrated to induce action errors.

C Comparison of TSR across different baseline adversarial attacks

We use OpenVLA as the baseline to evaluate MAVLA's stealthiness on two datasets. Perturbation features are generated and injected into the original images with $\sigma = 0.25$, and the attack effectiveness is compared against three baseline adversarial attacks. The evaluation focuses on TSR, the L_p -norm used to measure the perturbation magnitude and LPIPS employed to assess perceptual similarity. Smaller values of L_p -norm and LPIPS indicate stronger stealthiness. The results, summarized in Table 4, show that MAVLA consistently achieves comparable or superior attack effectiveness with the smallest perturbations, resulting in a significant drop in TSR. The LPIPS values before and after perturbation further validate MAVLA's stealthiness, showing that the generated perturbations

Table 4: Comparison of TSR Across Different Baseline Adversarial Attacks under The Same Perturbation Intensity

Datasets	Metrics	Baseline Attacks	Task 1	Task 2	Task 3	Task 4	Task 5	Task 6	Task 7	Task 8	Task 9	Task 10	Total
LIBERO-Spatial	TSR	Random Noise	90.1%	90.3%	76.2%	92.4%	62.2%	76.1%	94.2%	90.3%	82.0%	66.1%	81.9%
		Frequency Noise	76.4%	92.6%	36.5%	74.7%	60.4%	62.9%	80.2%	78.3%	84.1%	56.4%	70.3%
		Local Gaussian Mask	90.8%	88.7%	92.4%	90.1%	86.2%	88.3%	90.4%	94.7%	82.7%	64.5%	86.9%
		MAVLA	68.2%	73.8%	58.3%	92.7%	46.5%	64.6%	72.2%	18.9%	73.8%	21.7%	59.1%
	Lp -norm	Random Noise	38.790	38.800	38.800	38.800	38.790	38.790	38.790	38.790	38.790	38.790	38.793
		Frequency Noise	78.180	79.560	76.470	74.130	85.030	80.350	76.240	81.201	77.470	82.590	79.122
		Local Gaussian Mask	31.990	32.000	32.000	31.990	32.000	31.990	31.990	31.990	32.000	32.000	31.995
		MAVLA	4.470	4.401	4.480	4.486	4.459	4.510	4.554	4.515	4.410	4.452	4.474
	LPIPS	Random Noise	0.641	0.624	0.643	0.653	0.666	0.654	0.656	0.687	0.643	0.696	0.656
		Frequency Noise	0.167	0.168	0.169	0.172	0.183	0.173	0.172	0.194	0.173	0.176	0.175
		Local Gaussian Mask	0.052	0.053	0.054	0.054	0.056	0.055	0.055	0.056	0.053	0.056	0.054
		MAVLA	0.076	0.071	0.076	0.079	0.082	0.079	0.079	0.082	0.075	0.082	0.078
LIBERO-Goal	TSR	Random Noise	60.1%	92.0%	88.7%	18.4%	96.3%	92.7%	88.6%	99.4%	86.1%	60.5%	78.3%
		Frequency Noise	40.5%	58.4%	60.1%	24.6%	98.3%	72.6%	55.7%	78.6%	70.7%	38.9%	59.8%
		Local Gaussian Mask	56.9%	94.2%	84.8%	26.4%	98.3%	94.7%	98.2%	98.1%	82.5%	50.6%	78.5%
		MAVLA	46.3%	52.1%	66.6%	25.1%	88.1%	53.7%	52.1%	85.8%	72.2%	36.3%	57.8%
	Lp -norm	Random Noise	38.790	38.790	38.790	38.790	38.790	38.790	38.790	38.790	38.790	38.790	38.790
		Frequency Noise	73.530	77.150	75.420	76.400	77.040	73.390	73.530	77.150	75.420	76.400	75.543
		Local Gaussian Mask	31.990	31.990	31.990	31.990	31.990	31.990	31.990	32.000	32.000	32.000	31.993
		MAVLA	9.601	9.475	9.542	9.622	9.540	9.565	9.586	9.433	9.611	9.647	9.562
	LPIPS	Random Noise	0.648	0.650	0.624	0.631	0.647	0.642	0.643	0.651	0.629	0.955	0.672
		Frequency Noise	0.189	0.190	0.193	0.185	0.186	0.189	0.191	0.184	0.195	0.180	0.188
		Local Gaussian Mask	0.042	0.067	0.061	0.051	0.047	0.076	0.056	0.063	0.048	0.041	0.055
		MAVLA	0.130	0.131	0.132	0.135	0.133	0.127	0.133	0.129	0.132	0.130	0.131

Table 5: Computational and Deployment Cost of MAVLA

Models	LIBERO-Spatial					LIBERO-Goal				
	TTC / time step	EDL / s	PM / MiB	AGT / s/time step	MP/MB	TTC / time step	EDL / s	PM / MiB	AGT / s/time step	MP/MB
OpenVLA	128.36	-	30583	-	15370.2	139.61	-	30568	-	15370.2
OpenVLA+MAVLA	188.42	0.127	31091	0.0034	15462.3	210.8	0.1184	31020	0.0036	15462.3

are nearly imperceptible, with minimal perceptual differences from the original features in most cases. These findings confirm that MAVLA effectively balances stealthiness and attack strength while maintaining the performance.

D Computational and Deployment Cost of MAVLA

To assess MAVLA’s practical deployability, we measured its computational and deployment costs on OpenVLA using LIBERO-Spatial and LIBERO-Goal. We report four metrics: *TTC*, *EDL*, *PM*, and *AGT*. Results are summarized in Table 5. When MAVLA is activated, *TTC* increases notably compared to the only OpenVLA. The extra time steps stabilize at roughly 50%, reflecting repeated corrective attempts by the OpenVLA model under failure and further confirming the attack’s effectiveness. The one-time deployment cost is negligible ($EDL \leq 0.13s$), making its cost insignificant for multi-round or long-horizon tasks. *PM* usage rises by only about 1.5%, indicating a small incremental footprint for the perturbation generator. For sustained overhead, *AGT* remains below 4 ms/step, and the increase in model parameter count is about 0.6%. Overall, MAVLA achieves strong attack effectiveness and stealthiness with

controlled overhead, enabling both offline evaluation and near-real-time deployment in standard embodied agent environments.

E Discussion

While our work primarily exposes the vulnerabilities of VLA models, our findings also suggest potential defensive strategies. Since MAVLA relies on disrupting cross-modal alignment and injecting high-frequency perturbations to mislead the model, we posit that lightweight input transformations (e.g., minimal random resizing or high-fidelity image compression) can serve as an effective defense. Our analysis of the experimental results in Table 4 supports this hypothesis: the VLA model retains a relatively high task success rate (81.9%) under unstructured random noise, whereas the structured perturbations of MAVLA cause a significant drop to 59.1%. This performance gap indicates that VLA models are robust to benign, unstructured distortions but fragile to specific adversarial patterns. Therefore, by breaking the pixel-level dependency and sanitizing high-frequency components via input transformations, it is feasible to neutralize MAVLA with only a negligible compromise on benign task performance, providing a practical, training-free protection for embodied agents.